# Differential Dynamic Causal Nets: Model Construction, Identification and Group Comparisons —Supplementary Material

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## Supporting information

The supplementary material contains a Lie symmetry analysis of the canonical Jansen–Rit model (S1), an introduction to the JADE optimisation algorithm (S2), and permutation-based Wilcoxon signed-rank tests of NccDCM changes from the preictal to the ictal period (S3). It also describes bootstrap bias correction (S4), lagged correlation-based functional connectivity (S5), and EEG electrode allocation (S6). Further materials include post-ANOVA tables (S7), additional subject heterogeneity results (S8), bipolar EEG source localisation (S9), and regions of interest in the Desikan–Killiany atlas (S10). Finally, it presents the bifurcation behaviour of ccDCMs (S11), a three-node unidirectional ring network (S12), and the case in which a channel receives inputs from multiple other channels (S13).

## S1 A Lie symmetry analysis of the Jansen-Rit model

We introduce the reparameterisation

$$x_{A0} = x_0/A, B_A = A/B, C_A = AC, K_A = AK, \nu_0 = \nu_0.$$

The resulting model is

$$\begin{aligned} L_a x_{A0} &= aS(y), \\ L_a x_1 &= a\left(K_A z + 0.8 C_A S(C_A x_{A0})\right), \\ L_b x_2 &= 0.25 C_A b {B_A}^{-1} S(0.25 C_A x_{A0}), \\ y &= x_1 - x_2, \end{aligned} \tag{1}$$

where

$$\begin{aligned} L_a &= \partial^2/\partial t^2 + 2a\partial/\partial t^2 + a^2, \qquad L_b = \partial^2/\partial t^2 + 2b\partial/\partial t^2 + b^2, \\ S(v) &= \frac{\nu_{\max}}{1+\exp\left(r(v_0 - v)\right)}. \end{aligned}$$

The constants $a$, $b$, $e_0$ and $r$ and external input $z$ are assumed to be fixed, whereas $x_{A0}, x_1, x_2$ are hidden states and $\boldsymbol{\theta} = (B_A, C_A, K_A, \nu_0)$ are the unknown parameters.

For any continuous transformation $\phi(\cdot, \epsilon)$, if the output $y$ does not change when the parameter in (3) take values $\phi((B_A, C_A, K_A, \nu_0), \epsilon), 0 \le \epsilon \le \delta$. Then, we can find a small ball in the parametric space such that the output $y$ also does not change under the following transformation in a sufficiently small ball:

$$\begin{aligned} (x^*_{A0}, x^*_1, x^*_2, B^*_A, C^*_A, K^*_A, \nu^*_0) &= (\psi(x_{A0}, \epsilon_0), \psi(x_1, \epsilon_1), \psi(x_2, \epsilon_2), \psi(B_A, \epsilon_3), \\ &\quad \psi(C_A, \epsilon_4), \psi(K_A, \epsilon_5), \psi(\nu_0, \epsilon_6)), \end{aligned}$$

where

$$\psi(w, \epsilon) = \begin{cases} w + \epsilon, & w = 0 \\ \pm e^{\epsilon} w, & w \neq 0 \end{cases}$$

and $|\epsilon_i| \le \delta_i, i = 0, 1, ..., 6$. Let

$$\eta(w) = \partial\psi(w, \epsilon)/\partial\epsilon|_{\epsilon=0} = \begin{cases} 1, & w = 0 \\ \pm w, & w \neq 0 \end{cases}.$$

In general, suppose there exists a small parametric ball under which the output keeps the same for a time period. Then, we want to prove that the ball has zero radius.

For any inner point $(\theta, x_{A0}, x_1, x_2)$ in the parametric and state space, suppose that there exists a small ball in which for any $(x^*_{A0}, x^*_1, x^*_2, \theta^*)$, the output is preserved. We assume that there exist nonegative constants $\delta_i, i = 0, 1, ..., 6$, in which we can define a scaling map within the ball: $x^*_{A0} = e^{\epsilon_0} x_{A0}$, $x^*_1 = e^{\epsilon_1} x_1$ and $x^*_2 = e^{\epsilon_2} x_2$, $\theta^*_i = e^{\epsilon_{2+i}} \theta_i, i = 1, 2, 3, 4$, $|\epsilon_i| \le \delta_i$, $i = 0, 1, ..., 6$. To prove the local weak identifiability, it is sufficient to show that for the above Jansen-Rit model, only output-preserving scaling transformation is the identity mapping, that is, $\delta_i = 0, 1, ..., 6$.

Note that under the scaling transformations, the differential coefficients of $x_0, x_1, x_2, B_A, C_A, K_A, \nu_0$ are $\eta(x_{A0}) = \partial x^*_{A0}/\partial\epsilon|_{\epsilon=0} = x_{A0}$, $\eta(x_1) = x_1$, $\eta(x_2) = x_2$, $\eta(B_A) = B_A$, $\eta(C_A) = C_A$, $\eta(K_A) = K_A$ and $\eta(\nu_0) = \nu_0$. Following the Lie-symmetry formulation in Merkt et al. (2015), denote its infinitesimal generator by

$$\begin{aligned} X &= x_{A0} I(\delta_0 > 0) \frac{\partial}{\partial x_{A0}} + x_1 I(\delta_1 > 0) \frac{\partial}{\partial x_1} + x_2 I(\delta_2 > 0) \frac{\partial}{\partial x_2} + B_A I(\delta_3 > 0) \frac{\partial}{\partial B_A} \\ &\quad + C_A I(\delta_4 > 0) \frac{\partial}{\partial C_A} + K_A I(\delta_5 > 0) \frac{\partial}{\partial K_A} + \nu_0 I(\delta_6 > 0) \frac{\partial}{\partial \nu_0}, \end{aligned}$$

According to Theorem 2 in Merkt et al. (2015), the Lie symmetry preserves the output and, therefore, satisfies $X(y) = 0$ which gives

$$x_1 I(\delta_1 > 0) - x_2 I(\delta_2 > 0) = 0. \tag{2}$$

Following Yates et al. (2009), define the first prolonged function

$$F_1 = L_a x_{A0} - aS(y),$$

with the infinitesimal invariance condition

$$X(F_1) = 0$$

which gives

$$L_a(x_{A0})I(\delta_0 > 0) - S_v(y)(x_1 I(\delta_1 > 0) - x_2 I(\delta_2 > 0)) - aS_{v_0}(y)\nu_0 I(\delta_6 > 0) = 0. \tag{3}$$

Using the output condition (2), this reduces to

$$L_a(x_{A0})I(\delta_0 > 0) - aS_{v_0}(y)\nu_0 I(\delta_6 > 0) = 0.$$

Combining this with model (1) yields

$$aS(y)I(\delta_0 > 0) = aS_{v_0}(y)\nu_0 I(\delta_6 > 0) = -ar\nu_{\max} S(y)\left(1 - S(y)/\nu_{\max}\right)\nu_0 I(\delta_6 > 0)$$

as the derivatives of the sigmoid $S$ with respect to $\nu$ and $\nu_0$ satisfy

$$S_v(v) \widehat{=} \partial S(v)/\partial v = r\nu_{\max} S(v)\left(1 - S(v)/\nu_{\max}\right) = -\partial S(v)/\partial \nu_0 \widehat{=} - S_{\nu_0}(\nu).$$

We have

$$aI(\delta_0 > 0) = -ar\nu_{\max}\left(1 - S(y)/\nu_{\max}\right)\nu_0 I(\delta_6 > 0)$$

which implies $a(\delta_0 > 0) = 0$, $a\nu_0 I(\delta_6 > 0) = 0$ and the above equation implies

$$L_a(x_{A0})I(\delta_0 > 0) = 0. \tag{4}$$

Define the second prolonged function

$$F_2 = L_a x_1 - a\left(K_A z - 0.8 C_A S(C_A x_{A0})\right).$$

Note that

$$X(C_A x_{A0}) = x_{A0} C_A I(\delta_4 > 0) + C_A x_{A0} I(\delta_0 > 0).$$

Since $\nu_0 I(\delta_6 > 0) = 0$, the determining equation on the prolonged function

$$X(F_2) = 0$$

becomes

$$\begin{aligned} &L_a(x_1)I(\delta_1 > 0) - a\left(K_A z I(\delta_5 > 0) - 0.8 C_A S(C_A x_{A0}) C_A I(\delta_4 > 0)\right. \\ &\left. -0.8 C_A S_v(C_A x_{A0})\left(x_{A0} C_A I(\delta_4 > 0) + C_A x_{A0} I(\delta_0 > 0)\right)\right) = 0. \end{aligned} \tag{5}$$

Since $\delta_0 = 0$, combining the above determing equation with the model (1) yields

$$\begin{aligned} 0 &= a\left\{(K_A z + 0.8 C_A S(C_A x_{A0}))I(\delta_1 > 0)\right. \\ &\quad -K_A z I(\delta_5 > 0) + 0.8 C_A^2 S(C_A x_{A0}) I(\delta_4 > 0) \\ &\quad \left. +0.8 C_A S_\nu(C_A x_{A0})(x_{A0} C_A I(\delta_4 > 0) + C_A x_{A0} I(\delta_0 > 0))\right\} \\ &= a\left\{(K_A z + 0.8 C_A S(C_A x_{A0}))I(\delta_1 > 0)\right. \\ &\quad -K_A z I(\delta_5 > 0) + 0.8 C_A^2 S(C_A x_{A0}) I(\delta_4 > 0) \\ &\quad \left. +0.8 C_A^2 x_{A0} S_\nu(C_A x_{A0})(I(\delta_4 > 0) + I(\delta_0 > 0))\right\} \end{aligned}$$

which holds only if

$$aI(\delta_1 > 0) = 0, aC_A I(\delta_4 > 0) = 0, aK_A I(\delta_5 > 0) = 0.$$

Define the third prolonged function

$$F_3 = L_b x_2 - 0.25b\frac{C_A}{B_A}S(0.25C_A x_{A0})$$

$$X(F_3) = 0.$$

Note that

$$\begin{aligned} X\left(\frac{C_A}{B_A}\right) &= \frac{C_A}{B_A}I(\delta_4 > 0) - \frac{C_A}{B_A^2}B_A I(\delta_3 > 0) = -\frac{C_A}{B_A^2}B_A I(\delta_3 > 0), \\ X(0.25C_A x_{A0}) &= 0.25(x_{A0}C_A I(\delta_4 > 0) + C_A x_{A0} I(\delta_0 > 0)). \end{aligned}$$

Thus the determining equation on the prolonged function $F_3$ becomes

$$\begin{aligned} 0 &= L_b(x_2)I(\delta_2 > 0) + 0.25b\frac{C_A}{B_A^2}B_A I(\delta_3 > 0)S(0.25C_A x_{A0}) \\ &\quad -0.25bC_A {B_A}^{-1} S(0.25C_A x_{A0})0.25(x_{A0}C_A I(\delta_4 > 0) + C_A x_{A0} I(\delta_0 > 0)) \\ &= 0.25bC_A {B_A}^{-1} S(0.25C_A x_{A0})(I(\delta_2 > 0) + I(\delta_3 > 0)) \\ &\quad -0.25^2 bC_A {B_A}^{-1} S(0.25C_A x_{A0})(x_{A0}C_A I(\delta_4 > 0) + C_A x_{A0} I(\delta_0 > 0)) \\ &= 0.25bC_A {B_A}^{-1} S(0.25C_A x_{A0}) \\ &\quad \times (I(\delta_2 > 0) + I(\delta_3 > 0) - 0.25x_{A0}C_A(I(\delta_4 > 0) + I(\delta_0 > 0))) \end{aligned}$$

which yields

$$0 = bC_A(I(\delta_2 > 0), 0 = bC_A(I(\delta_3 > 0) = 0, bC_A I(\delta_4 > 0)) = 0, bC_A I(\delta_0 > 0) = 0. \quad (6)$$

Combining the above two equations give

$$L_b(x_2)I(\delta_2 > 0) = 0. \quad (7)$$

The output-preserving condition

$$x_1 I(\delta_1 > 0) = x_2 I(\delta_2 > 0).$$

gives $L_a(x_2)I(\delta_2 > 0) = L_a(x_1)I(\delta_1 > 0) = 0$.

Let $D_t = \partial/\partial t$. subtracting $L_b$ from $L_a$, on $\delta_2 > 0$, we have

$$0 = (La - L_b)(x_2) = 2(a - b)D_t x_2 + (a^2 - b^2)x_2.$$

For $a \neq b$,

$$2D_t x_2 + (a + b)x_2 = 0$$

which yields

$$x_2(t) = c\exp(-(a + b)t/2).$$

Substitution $x_2(t)$ into $L_a(x_2) = 0$ gives

$$\frac{(a - b)^2}{4}c = 0.$$

Hence, for $a \neq b$,

$$x_2(t) = 0 \tag{8}$$

which implies

$$0 = L_b(x_)2 = 0.25 C_A b {B_A}^{-1} S(0.25 C_A x_{A0}) \, .$$

On $(\delta_2 > 0)$, $bC_A = 0$. Similarly, on $(\delta_1 > 0)$, $aC_A = 0$ and $x_1(t) = 0$.

In summary, for $a \neq b$ and non-zero $B_A, C_A, K_A, \nu_0$, we have $\delta_i = 0, i = 0, 1, ..., 6$. Therefore, there is no nontrivial continuous Lie transformation acting on the parameter vector

$$(B_A, C_A, K_A, v_0)$$

while preserving the output.

For the cases that $a \not\sim b$ or $a = 0$, the differential equations in (1) have no solutions.

Consequently, parameters $(B_A, C_A, K_A, v_0)$ have no non-trival Lie symmetry for the canonical Jansen-Rit model.

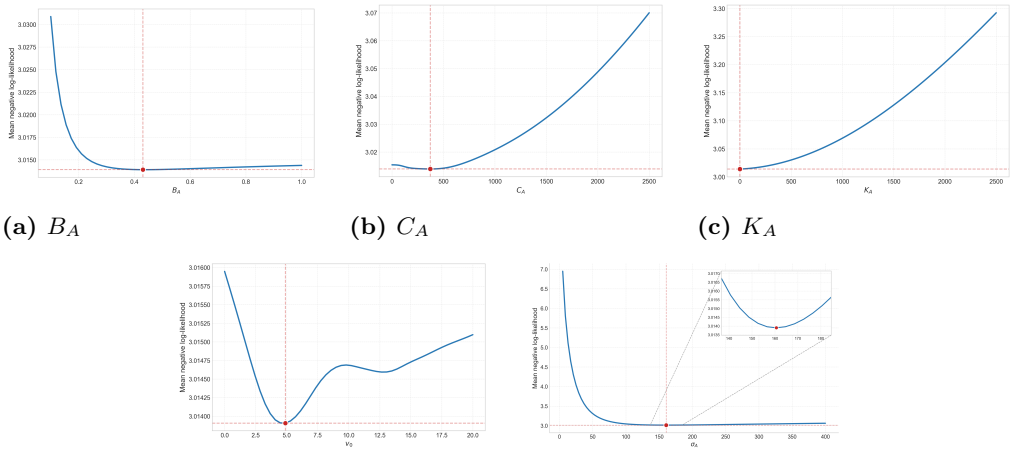
## S2 The optimisation algorithm JADE

To minimize the forward loss function, we employed an adaptive differential evolution with optional external archive (JADE), a population-based global optimization method proposed by Zhang and Sanderson (2009) and implemented in the `mealpy` library. JADE is well suited for non-convex, multi-modal, and non-differentiable optimization problems, as it does not rely on gradient information. Instead, it evolves a population of candidate solutions through mutation, crossover, and selection, with the aim of converging to the global minimum. The method is particularly advantageous in our context because the objective function may be non-smooth or contain multiple local minima due to the models nonlinear structure.

As a metaheuristic optimization algorithm, the performance of JADE is sensitive to the choice of hyperparameters, the initial population size and epoch size. These two hyperparameters substantially affect both the convergence speed and the quality of the final solution.

Figure S2 illustrates how the value of the loss function evolves under different settings of population size and iteration number in simulation studies, emphasizing the role of these hyperparameters in shaping optimisation performance. It is evident that almost algorithm JADE eventually converges to the same optimal value when the number of iterations exceeds 150 for almost all population sizes. Intuitively, smaller populations may force the algorithm to converge to a local minimum, limiting the coverage of its search space. In contrast, larger populations generally facilitate faster convergence owing to enhanced diversity, though further increases in population size yield diminishing returns in performance while incurring substantially higher computational costs. To strike a balance between convergence accuracy and computational efficiency, we set the population size and the maximum number of iterations to 15 and 250, respectively.

We assessed the sensitivity of the JADE-based optimisation by fitting the NccDCM to the EEG data from subject P3, obtaining the parameter estimates $(\widehat{B}_A, \widehat{C}_A, \widehat{K}_A, \widehat{v}_0, \widehat{\sigma}_A)$. For each parameter in turn, we evaluated the loss function while holding the other four parameters fixed at their estimated values. The resulting one-dimensional loss profiles (Figure S1) appear locally convex in the vicinity of the estimated parameter values, although several local minima are also evident.



**(a)** $B_A$ **(b)** $C_A$ **(c)** $K_A$

**(d)** $v_0$ **(e)** $\sigma_A$

**Figure S1.** Plots of loss function in $B_A$, $C_A$, $K_A$, $v_0$, and $\sigma_A$.

# S3 Multiple permutation-based Wilcoxon signed-rank tests for significant NccDCM changes from the preictal period to the ictal period

For each of parameters in $(B_{Anq}^{(ilj)k}, C_{Anq}^{(i|j)k}, K_{Anq}^{(i|j)k}, v_{0nq}^{(i|j)k}, \sigma_{Anq}^{(i)k})$ and for each node pair, let $\Pi_{\text{pre}}^{(\cdot)}$ and $\Pi_{\text{on}}^{(\cdot)}$ denote the matrices of estimates derived from the NccDCM in the preictal and ictal periods respectively. Column of $\Pi_{\text{pre}}^{(\cdot)}$ and $\Pi_{\text{on}}^{(\cdot)}$ stand for the estimators of different case-subjects indexed by $1 \leq n \leq N_1$ while rows correspond to different time segments indexed by $1 \leq q \leq 6$. We performed multiple permutation-based Wilcoxon signed-rank tests for significant NccDCM changes from the preictal period to the ictal period They are implemented in four steps as follows.

- **Compute the test statistics** $W = (W_1, W_2, \ldots, W_{N_1})^\top$**.** For subject $n$, denote the $n$-th columns of $\Pi_{\text{pre}}^{(\cdot)}$ and $\Pi_{\text{on}}^{(\cdot)}$ as $\pi_{\text{pre}}^{(\cdot,j)}$ and $\pi_{\text{on}}^{(\cdot,j)}$, respectively. Compute the paired differences $\pi_{\text{on}}^{(\cdot,n)} - \pi_{\text{pre}}^{(\cdot,n)}$. After removing zero differences, rank the absolute values of the remaining differences in ascending order, assigning average ranks in the case of ties. The statistic $W_n$ is then obtained by summing the ranks corresponding to positive differences, i.e., those instances where $\pi_{\text{on}}^{(\cdot,n)} > \pi_{\text{pre}}^{(\cdot,n)}$.

- **Pair-matched permutation procedure.** For each column in $\Pi_{\text{pre}}^{(\cdot)}$, find the corresponding column in $\Pi_{\text{on}}^{(\cdot)}$. We randomly swap the corresponding entries in the two columns, yielding $2^6$ distinct permutations. For each permutation $\alpha$, we denote the resulted permuted matrices by $\Pi_{\text{pre}}^{(\cdot),\alpha}$ and $\Pi_{\text{on}}^{(\cdot),\alpha}$. $W^\alpha$ is reduced to the observed statistic $W$ when $\alpha$ is an identity permutation. The procedure is implemented as follows.

- **Compute the empirical** $p$**-values.** For the $n$-th subject, the empirical $p$-value is computed as

$$p^{(n)} = \frac{\#\{W_n^\alpha \geq W_n\}}{\text{Total number of permutations}},$$

  where the numerator counts the number of permuted statistics at least as extreme as the observed $W_n$. This yields the individual $p$-value for each subject.

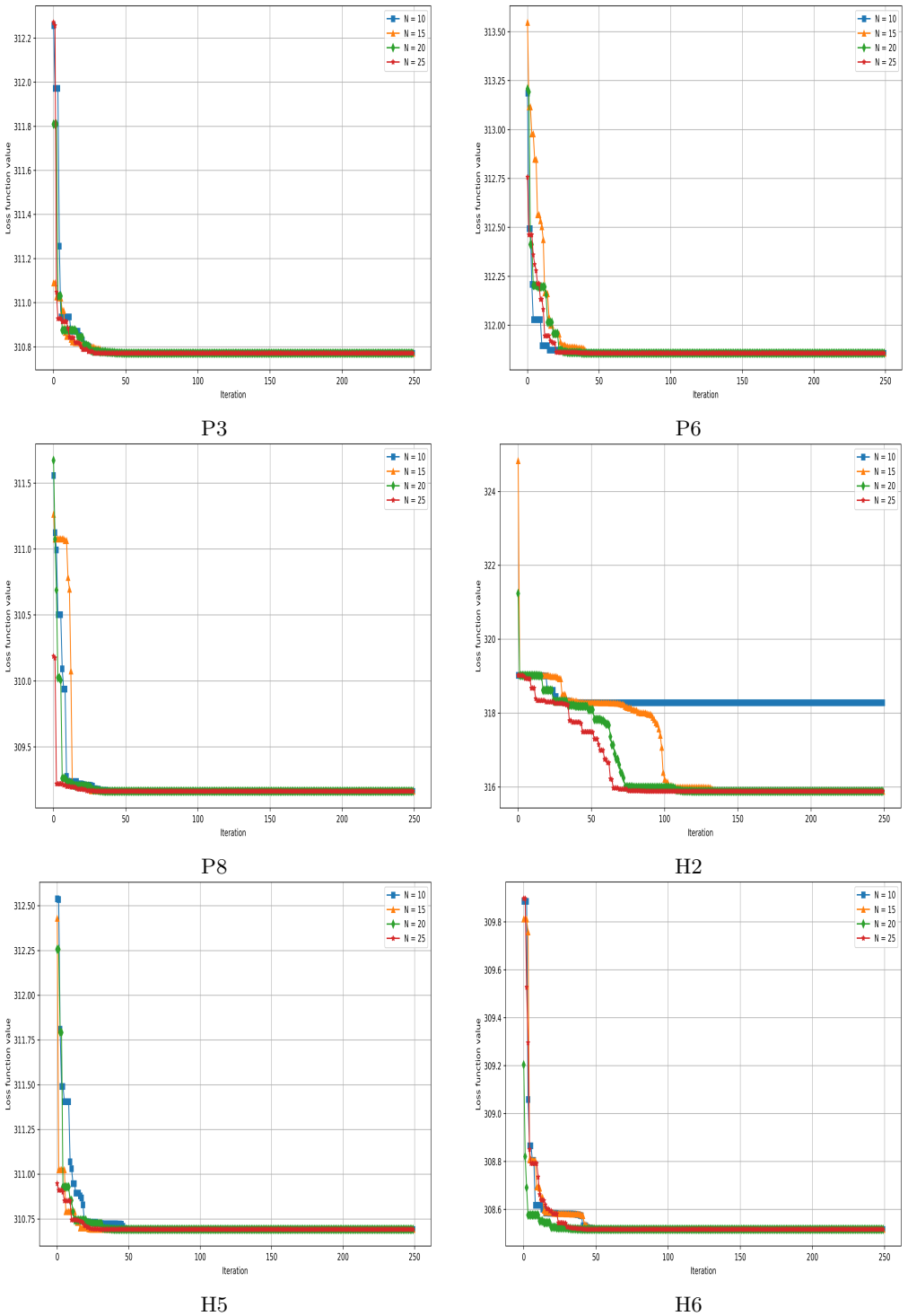



**Figure S2.** Examples of trace plots of the loss function as the number of iterations is increasing for subjects P3, P6, P8, H2, H5 and H6: L-Shape under different population sizes.

- **Combine the $p$-values using Fisher's method.** Combining these individual

$p$-values using Fisher's method, we have the combined test statistic

$$\chi^2_{\text{Fisher}} = -2\sum_{n=1}^{N_1} \ln p^{(n)}.$$

The above statistic follows a chi-square distribution with $2 \times N_1$ degrees of freedom under the null hypothesis that there are no changes in the parameters from the preictal period to the ictal period. The overall significance can derived from the above chi-square distribution.

Analogous to the case-control group comparison, adjustment for multiple input effects can be also implemented. The details are omitted.

# S4 Bootstrap bias correction

The detailed parameter estimation procedure is described in the following steps.

**Step 1** Apply the Chen–Fliess method to construct the objective function (**see Section 2.2**).

**Step 2** The parameters are estimated using the genetic evolutionary algorithm JADE, yielding the estimator $\widehat{\theta} = (\widehat{B}_A,\ \widehat{C}_A,\ \widehat{K}_A,\ \widehat{v}_0,\ \widehat{\sigma}_A)$.

**Step 3** Based on the estimated parameter values $\widehat{\theta}$ and the canonical Jansen–Rit model **(4)**, we generate $n_{\text{boot}}$ synthetic EEG signals and use these signals to obtain $n_{\text{boot}}$ bootstrap estimators $\widehat{\theta}^{*,b}, b = 1, \ldots, n_{\text{boot}}$.

Define the bootstrap mean estimator by $\widehat{\theta}^* = \frac{1}{n_{\text{boot}}} \sum_{b=1}^{n_{\text{boot}}} \widehat{\theta}^{*,b}$. Then, the bias-corrected estimator is given by $\widehat{\theta}_B = 2\widehat{\theta} - \widehat{\theta}^*$.

Note that, during the bootstrap procedure, since the true parameter values used to generate the synthetic EEG signals are known, we employ the `minimize` function provided in `scipy` for parameter estimation. Moreover, the initial value of the optimization algorithm is set to $\widehat{\theta}$ to improve computational efficiency.

# S5 Lagged correlation-based functional connectivity

We calculate the functional connectivity based on the lagged correlation coefficient. The EEG signals used are extracted from the same time block across all subjects and from all channels. For each subject, we compute pairwise cross-correlations between all channel pairs across time lags ranging from $-50$ to $+50$. For each channel pair, the maximum absolute correlation across these time lags is calculated, which allows us to construct correlation matrices for all subjects.

To normalise the correlation values, we apply Fisher's $z$-transform, yielding the matrices $F_{P_i}$ $(1 \le i \le n_0)$ and $F_{H_j}$ $(1 \le j \le n_1)$. For the control and case groups, the mean matrices are then computed as $\overline{F}_P = \frac{1}{n_0}\sum_{i=1}^{n_0} F_{P_i}$, $\overline{F}_H = \frac{1}{n_1}\sum_{j=1}^{n_1} F_{H_j}$. Then, the corresponding variance matrices are defined as $\overline{V}_P = \frac{1}{n_0}\sum_{i=1}^{n_0}\big(F_{P_i} - \overline{F}_P\big)^2$ and $\overline{V}_H = \frac{1}{n_1}\sum_{i=1}^{n_1}\big(F_{H_i} - \overline{F}_H\big)^2$. Let $\overline{\rho}_{ij}^{(0)}$, $\overline{\rho}_{ij}^{(1)}$, $\overline{\varrho}_{ij}^{(0)}$, and $\overline{\varrho}_{ij}^{(1)}$ denote the $(i,j)$-th elements of $\overline{F}_P$, $\overline{F}_H$, $\overline{V}_P$, and $\overline{V}_H$, respectively. For each channel pair, we compute the z-score as

$$f_{ij} = \frac{\overline{\rho}_{ij}^{(0)} - \overline{\rho}_{ij}^{(1)}}{\sqrt{\dfrac{(n_0-1)\,\overline{\varrho}_{ij}^{(0)} + (n_1-1)\,\overline{\varrho}_{ij}^{(1)}}{n_0+n_1-2}}}.$$

Finally, only the values of $|f_{ij}|$ greater than 0.5 are retained, and the results are based on the absolute z-scores.

In addition, we apply the following permutation test procedure to compute the $p$-values.

1. **Permutation procedure.** Let $\mathcal{D} = (H_1, \ldots, H_{10}, P_1, P_3, P_5, \ldots, P_{10})$ denote the combined dataset. To assess the statistical significance of the observed $F = (f_{ij})$, we apply a permutation test: the labels "case" and "control" are randomly reassigned by permuting $\mathcal{D}$. In each permutation, the first 10 subjects in the permuted dataset $\mathcal{D}^\pi$ are designated as the control group and the remaining subjects as the case group. Then, we recompute the test statistic $f_{ij}^\pi$.

2. **Significance assessment.** This permutation procedure is repeated 999 times. The observed statistic $f_{ij}$ is included as part of the permutation distribution $\{f_{ij}^\pi\}$. The empirical $p$-value is then given by

$$p = \frac{\#\{|f_{ij}^\pi| \geq |f_{ij}|\}}{\text{Total number of permutations}},$$

   where the numerator counts the number of permuted statistics at least as extreme as the observed $f_{ij}$.

## S6 EEG electrode allocation on the scalp

Table S1 shows the positions of the electrodes and the brain regions that the channels cover.

**Table S1.** EEG electrode pairs and their brain regions

| Electrode pair | Position | Brain Region |
|---|---|---|
| FP2F4 | Right frontopolar | Right prefrontal cortex |
| P4O2 | Right parieto-occipital | Visual association |
| F4C4 | Right fronto-central | Right frontal → motor cortex |
| F3C3 | Left fronto-central | Left frontal → motor cortex |
| C4P4 | Right centro-parietal | Right sensorimotor → parietal |
| C3P3 | Left centro-parietal | Left sensorimotor → parietal |
| FP1F3 | Left frontopolar | Left prefrontal cortex |
| P3O1 | Left parieto to occipital | Visual cortex |
| FP2F8 | Right frontopolar | Anterior inferior frontal |
| F8T4 | Right inferior frontal | Broca homolog |
| T4T6 | Right mid to posterior temporal | Auditory association |
| T6O2 | Right posterior temporal | Visual association |
| FP1F7 | Left frontopolar | Anterior inferior frontal cortex |
| F7T3 | Left inferior frontal | Broca area vicinity |
| T3T5 | Left mid to posterior temporal lobe | Auditory and language |
| T5O1 | Left posterior temporal to occipital | Visual association |
| FZCZ | Medial frontal cortex | Supplementary motor |
| CZPZ | Central midline to parietal midline | Primary motor |
| BP3REF | A reference electrode | Not a cortical region |

**Table S2.** Post ANOVA: $B_A$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 361.3590 | 0.0471 | 268.4587 | 0.0002 |
| Channel | 305 | 405.1509 | 0.0528 | 0.9869 | 0.5942 |
| Case-control:Channel | 305 | 311.8932 | 0.0407 | 0.7597 | 1.0000 |
| Residual | 4896 | 6590.2635 | 0.8594 | – | – |
| Total | 5507 | 7668.6666 | 1.0000 | – | – |

**Table S3.** Post ANOVA: $C_A$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 505865346.0343 | 0.0191 | 108.8468 | 0.0002 |
| Channel | 305 | 2003473690.6040 | 0.0757 | 1.4134 | 0.0002 |
| Case-control:Channel | 305 | 1186482323.8943 | 0.0449 | 0.8370 | 1.0000 |
| Residual | 4896 | 22754144867.7666 | 0.8603 | – | – |
| Total | 5507 | 26449966228.2993 | 1.0000 | – | – |

# S7 Post-ANOVA tables

The results of the post ANOVA are summarised in Tables S2—S11. The results indicated that the identified dynamic causal nets were highly significant for each parameter.

# S8 More results on subject-heterogeneity analysis

We present results of subject-heterogeneity analysis for FP1F7, C4P4, T6O2, T5O1 and P4O2. The results are displayed in Figure S3.

# S9 Bipolar EEG source localisation

*EEG Data Acquisition and Preprocessing.* Electroencephalography (EEG) recordings were obtained for 11 subjects in European Data Format (EDF). Signals were recorded using a clinical bipolar montage comprising 18 derivations (Fp2–F4, F4–C4, C4–P4, P4–O2, Fp1–F3, F3–C3, C3–P3, P3–O1, Fp2–F8, F8–T4, T4–T6, T6–O2, Fp1–F7, F7–T3, T3–T5, T5–O1, Fz–Cz, Cz–Pz). A non-EEG channel BP3–REF was excluded from further analysis. EEG signals were band-pass filtered between 1–40 Hz and re-referenced implicitly through the bipolar montage. No additional re-referencing was applied to preserve the original spatial derivations. Independent component analysis was applied and six components were chosen. Artefacts due to eye-movements and blink were removed. Segmentation information provided in accompanying metadata identified

**Table S4.** Post ANOVA: $K_A$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 102982330.7693 | 0.0150 | 104.0527 | 0.0002 |
| Channel | 305 | 1469172393.7312 | 0.2144 | 4.8670 | 0.0002 |
| Case-control:Channel | 305 | 434145625.5117 | 0.0634 | 1.4382 | 0.0002 |
| Residual | 4896 | 4845636908.6169 | 0.7072 | – | – |
| Total | 5507 | 6851937258.6290 | 1.0000 | – | – |

**Table S5.** Post ANOVA: $\nu_0$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 184766.4927 | 0.0296 | 168.7859 | 0.0002 |
| Channel | 305 | 426044.4232 | 0.0684 | 1.2761 | 0.0002 |
| Case-control:Channel | 305 | 262557.1241 | 0.0421 | 0.7864 | 1.0000 |
| Residual | 4896 | 5359552.0694 | 0.8599 | – | – |
| Total | 5507 | 6232920.1094 | 1.0000 | – | – |

**Table S6.** Post ANOVA: $\sigma_A$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 15530065.7022 | 0.1430 | 873.5486 | 0.0002 |
| Channel | 305 | 4757543.9657 | 0.0438 | 0.8774 | 0.9592 |
| Case-control:Channel | 305 | 1287117.7224 | 0.0119 | 0.2374 | 1.0000 |
| Residual | 4896 | 87041755.7707 | 0.8014 | – | – |
| Total | 5507 | 108616483.1611 | 1.0000 | – | – |

**Table S7.** Source Post ANOVA: $B_A$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 181.7473 | 0.0226 | 99.5304 | 0.0002 |
| Cortical region | 235 | 595.5941 | 0.0740 | 1.3879 | 0.0002 |
| Case-control:Cortical region | 235 | 377.3340 | 0.0469 | 0.8793 | 0.9837 |
| Residual | 3776 | 6895.1545 | 0.8566 | – | – |
| Total | 4247 | 8049.8298 | 1.0000 | – | – |

**Table S8.** Source Post ANOVA: $C_A$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 640100846.1766 | 0.0535 | 258.3829 | 0.0002 |
| Cortical region | 235 | 1228835491.5024 | 0.1026 | 2.1108 | 0.0002 |
| Case-control:Cortical region | 235 | 750632429.7321 | 0.0627 | 1.2894 | 0.0002 |
| Residual | 3776 | 9354414664.0195 | 0.7812 | – | – |
| Total | 4247 | 11973983431.4305 | 1.0000 | – | – |

**Table S9.** Source Post ANOVA: $K_A$.

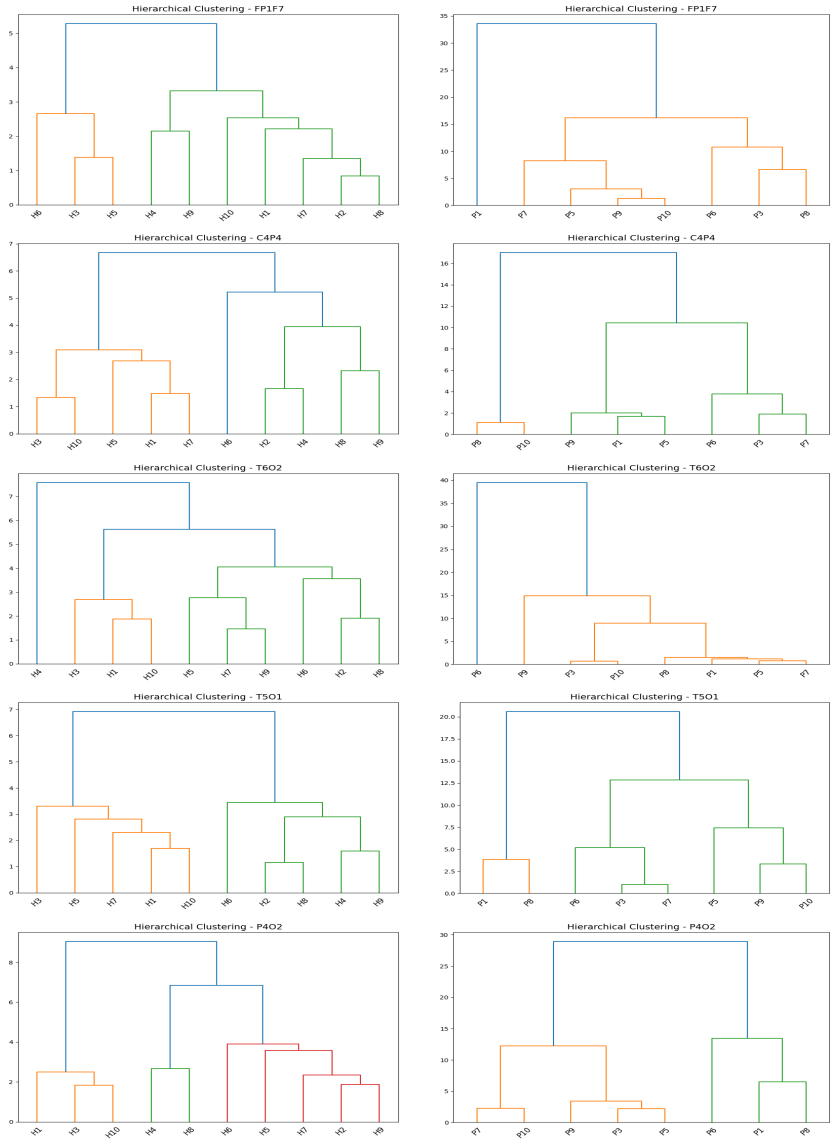
| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 7227433.4849 | 0.0018 | 12.2902 | 0.0003 |
| Cortical region | 235 | 1539002062.1353 | 0.3869 | 11.1364 | 0.0002 |
| Case-control:Cortical region | 235 | 211223240.4413 | 0.0531 | 1.5284 | 0.0002 |
| Residual | 3776 | 2220533084.5214 | 0.5582 | – | – |
| Total | 4247 | 3977985820.5830 | 1.0000 | – | – |

**Table S10.** Source Post ANOVA: $\nu_0$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 132730.9279 | 0.0267 | 122.2962 | 0.0002 |
| Cortical region | 235 | 420670.2229 | 0.0847 | 1.6494 | 0.0002 |
| Case-control:Cortical region | 235 | 315820.1184 | 0.0636 | 1.2383 | 0.0002 |
| Residual | 3776 | 4098179.8733 | 0.8250 | – | – |
| Total | 4247 | 4967401.1424 | 1.0000 | – | – |

**Table S11.** Source Post ANOVA: $\sigma_A$.

| Factor | Df | Sum of Sq. | $R^2$ | F | p-value |
|---|---|---|---|---|---|
| Case-control | 1 | 7298611.6045 | 0.0185 | 103.2300 | 0.0002 |
| Cortical region | 235 | 114634103.3495 | 0.2901 | 6.8994 | 0.0002 |
| Case-control:Cortical region | 235 | 6276374.4906 | 0.0159 | 0.3778 | 1.0000 |
| Residual | 3776 | 266972371.2363 | 0.6756 | – | – |
| Total | 4247 | 395181460.6809 | 1.0000 | – | – |




Controls Cases

**Figure S3.** Dendrograms for the channels FP1F7, C4P4, T6O2, T5O1 and P4O2. The left panel is for the control group (the left panel) while the right panel is for the case group.

seizure (ictal) intervals via start and end times; all remaining intervals were treated as non-seizure (interictal) periods.

*Segmentation of EEG Data.* For each subject, seizure segments were extracted based on annotated start–end times. Non-seizure segments were defined as the complementary intervals of the continuous recording. Segments shorter than 5 seconds were excluded to ensure reliable estimation of temporal statistics. To mitigate boundary artifacts, 0.5 seconds were trimmed from the beginning and end of each segment.

*Forward Model and Source Space.* EEG source localisation was performed using a template-based approach implemented in MNE-Python. A standard anatomical template (fsaverage) was used in the absence of subject-specific MRI data. The cortical source space was defined using a surface-based discretization ( 5,000–10,000 vertices per hemisphere). A three-layer boundary element model (BEM) was constructed to model volume conduction. The standard forward model is defined as:

$$B = LQ + E$$

where $B$ denotes monopolar EEG measurements, $L$ the leadfield matrix, $Q$ the cortical source activity, and $E$ measurement noise.

*Bipolar Montage Transformation.* Because EEG data were recorded in a bipolar montage, the measured signals $A$ were related to the underlying monopolar potentials $B$ through a linear transformation:

$$A = MB$$

where $M$ is a binary-valued link matrix encoding electrode differences (entries in $\{-1, 0, 1\}$). Substituting the forward model yields:

$$A = (ML)Q + ME$$

Thus, source localisation was performed by replacing the standard leadfield $L$ with the transformed leadfield $ML$, ensuring consistency between the measurement model and the recorded data. The noise covariance matrix was transformed accordingly. The marginal covariances of $A$, $B$, $Q$ and $E$ satisfy:

$$\mathrm{cov}(A) = M\mathrm{cov}(B)M^T = ML\mathrm{cov}(Q)L^TM^T + M\mathrm{cov}(E)M^T.$$

*Source Reconstruction.* Distributed source reconstruction was performed using minimum-norm estimation (MNE) with depth weighting and noise normalization (dSPM). The inverse operator was computed using the transformed leadfield $ML$ and corresponding noise covariance. Source time series were estimated for each segment independently. Let $M = UDV^T$ denote the singular value decomposition and $M^+ = VD^+U^T$ the pseudoinverse (i.e., the Moore–Penrose inverse) of $M$. Let $\Sigma$ denote $M\mathrm{cov}(E)M^T$. The minimum norm estimate is defined by

$$\begin{aligned}\widehat{Q} &= \arg\min_Q (A - MLQ)^T(A - MLQ) + \lambda\Sigma^{1/2}QQ^T\lambda\Sigma^{1/2} \\ &= ((L^TM^T\Sigma^+ML + \lambda I)^{-1}L^TM^T\Sigma^+A.\end{aligned}$$

*Cortical Parcellation.* Reconstructed source activity was mapped onto 68 cortical regions defined by the Desikan–Killiany atlas. For each region, vertex-wise time series were aggregated using sign-flipped means to average the estimated source time series over the grid points within each region, yielding one representative time series per region.

## S10 Region of Interest (ROI): Desikan-Killiany atlas

**Table S12.** Cortical region indices, region names, and abbreviations.

| Index | Region Name | Abbreviation |
|---|---|---|
| 1 | ctx-lh-bankssts | lh-BSTS |
| 2 | ctx-lh-caudalanteriorcingulate | lh-cACC |
| 3 | ctx-lh-caudalmiddlefrontal | lh-CMF |
| 4 | ctx-lh-cuneus | lh-CUN |
| 5 | ctx-lh-entorhinal | lh-Ent |
| 6 | ctx-lh-frontalpole | lh-FP |
| 7 | ctx-lh-fusiform | lh-Fus |
| 8 | ctx-lh-inferiorparietal | lh-IP |
| 9 | ctx-lh-inferiortemporal | lh-IT |
| 10 | ctx-lh-insula | lh-Ins |
| 11 | ctx-lh-isthmuscingulate | lh-ICC |
| 12 | ctx-lh-lateraloccipital | lh-LO |
| 13 | ctx-lh-lateralorbitofrontal | lh-LOF |
| 14 | ctx-lh-lingual | lh-Ling |
| 15 | ctx-lh-medialorbitofrontal | lh-MOF |
| 16 | ctx-lh-middletemporal | lh-MT |
| 17 | ctx-lh-paracentral | lh-ParC |
| 18 | ctx-lh-parahippocampal | lh-PH |
| 19 | ctx-lh-parsopercularis | lh-POp |
| 20 | ctx-lh-parsorbitalis | lh-POr |
| 21 | ctx-lh-parstriangularis | lh-PTr |
| 22 | ctx-lh-pericalcarine | lh-Cal |
| 23 | ctx-lh-postcentral | lh-PoC |
| 24 | ctx-lh-posteriorcingulate | lh-PCC |
| 25 | ctx-lh-precentral | lh-PC |
| 26 | ctx-lh-precuneus | lh-PreCu |
| 27 | ctx-lh-rostralanteriorcingulate | lh-rACC |
| 28 | ctx-lh-rostralmiddlefrontal | lh-RMF |
| 29 | ctx-lh-superiorfrontal | lh-SF |
| 30 | ctx-lh-superiorparietal | lh-SP |
| 31 | ctx-lh-superiortemporal | lh-ST |
| 32 | ctx-lh-supramarginal | lh-SM |
| 33 | ctx-lh-temporalpole | lh-TP |
| 34 | ctx-lh-transversetemporal | lh-TT |
| 35 | ctx-rh-bankssts | rh-BSTS |
| 36 | ctx-rh-caudalanteriorcingulate | rh-cACC |
| 37 | ctx-rh-caudalmiddlefrontal | rh-CMF |
| 38 | ctx-rh-cuneus | rh-CUN |
| 39 | ctx-rh-entorhinal | rh-Ent |
| 40 | ctx-rh-frontalpole | rh-FP |
| 41 | ctx-rh-fusiform | rh-Fus |
| 42 | ctx-rh-inferiorparietal | rh-IP |
| 43 | ctx-rh-inferiortemporal | rh-IT |
| 44 | ctx-rh-insula | rh-Ins |
| 45 | ctx-rh-isthmuscingulate | rh-ICC |
| 46 | ctx-rh-lateraloccipital | rh-LO |

Continued on next page

| Index | Region Name | Abbreviation |
|---|---|---|
| 47 | ctx-rh-lateralorbitofrontal | rh-LOF |
| 48 | ctx-rh-lingual | rh-Ling |
| 49 | ctx-rh-medialorbitofrontal | rh-MOF |
| 50 | ctx-rh-middletemporal | rh-MT |
| 51 | ctx-rh-paracentral | rh-ParC |
| 52 | ctx-rh-parahippocampal | rh-PH |
| 53 | ctx-rh-parsopercularis | rh-POp |
| 54 | ctx-rh-parsorbitalis | rh-POr |
| 55 | ctx-rh-parstriangularis | rh-PTr |
| 56 | ctx-rh-pericalcarine | rh-PCal |
| 57 | ctx-rh-postcentral | rh-PoC |
| 58 | ctx-rh-posteriorcingulate | rh-PCC |
| 59 | ctx-rh-precentral | rh-PC |
| 60 | ctx-rh-precuneus | rh-PreC |
| 61 | ctx-rh-rostralanteriorcingulate | rh-rACC |
| 62 | ctx-rh-rostralmiddlefrontal | rh-RMF |
| 63 | ctx-rh-superiorfrontal | rh-SF |
| 64 | ctx-rh-superiorparietal | rh-SP |
| 65 | ctx-rh-superiortemporal | rh-ST |
| 66 | ctx-rh-supramarginal | rh-SM |
| 67 | ctx-rh-temporalpole | rh-TP |
| 68 | ctx-rh-transversetemporal | rh-TT |

## S11 Bifurcation behavior of ccDCMs

As ccDCMs play a key role in shaping and influencing the structure and behavior of the NccDCM, in the following we exam the behavior of ccDCM and evaluate the accuracy of the proposed parameter estimation. For this purpose, suppressing channel and group subscripts in the model (4), consider the following ccDCM with states $\boldsymbol{x}(t) = (x_0(t), x_1(t), x_2(t), x_3(t), x_4(t), x_5(t))$:

$$\begin{aligned} dx_0(t) &= x_3(t)dt, \quad dx_1(t) = x_4(t)dt, \quad dx_2(t) = x_5(t)dt, \\ dx_3(t) &= \left[Aa\mathrm{S}\left(x_1(t) - x_2(t)\right) - 2ax_3(t) - a^2x_0(t)\right]dt, \\ dx_4(t) &= \left[Aa\left\{0.8C\mathrm{S}\left(Cx_0(t)\right) + K\mathrm{S}\left(y_2(t)\right)\right\} - 2ax_4(t) - a^2x_1(t)\right]dt + Aadw(t), \\ dx_5(t) &= \left[Bb0.25C\mathrm{S}\left(0.25Cx_0(t)\right) - 2bx_5(t) - b^2x_2(t)\right]dt, \end{aligned} \tag{9}$$

with initial value $\boldsymbol{x}_0 = (x_0(0), ..., x_5(0))^T \in \mathbb{R}^6$ and the input $y_2(t)$ from the external channel. We use the following Strang splitting method to generate the samples of $\boldsymbol{x}(t)$ from the above model. See (Milstein and Tretyakov, 2003; Leimkuhler et al., 2016) for more details. To this end, we first divide the system (9) into two subsystems below:

$$\begin{aligned} dx_0^{[1]}(t) &= x_3^{[1]}(t)dt, \quad dx_1^{[1]}(t) = x_4^{[1]}(t)dt, \quad dx_2^{[1]}(t) = x_5^{[1]}(t)dt, \\ dx_3^{[1]}(t) &= [-2ax_3^{[1]}(t) - a^2x_0^{[1]}(t)]dt, \\ dx_4^{[1]}(t) &= \left[-2ax_4^{[1]}(t) - a^2x_1^{[1]}(t)\right]dt, \\ dx_5^{[1]}(t) &= [-2bx_5^{[1]}(t) - b^2x_2^{[1]}(t)]dt, \end{aligned} \tag{10}$$

and

$$\begin{aligned} dx_0^{[2]}(t) &= 0, \quad dx_1^{[2]}(t) = 0, \quad dx_2^{[2]}(t) = 0, \\ dx_3^{[2]}(t) &= Aa\mathrm{S}\left(x_1^{[2]}(t) - x_2^{[2]}(t)\right)dt, \\ dx_4^{[2]}(t) &= Aa\left[0.8C\mathrm{S}\left(Cx_0^{[2]}(t)\right) + K\mathrm{S}\left(y_2(t)\right)\right]dt + Aadw(t), \\ dx_5^{[2]}(t) &= Bb0.25C\mathrm{S}\left(0.25Cx_0^{[2]}(t)\right)dt. \end{aligned} \tag{11}$$

Let $\varphi_t^{[1]}$ and $\varphi_t^{[2]}$ denote the exact flows of subsystems (10) and (11). Then, for $\boldsymbol{x} \in \mathbb{R}^6$, the so-called Strang splitting for system (9) can be written as

$$\psi_{\Delta t}(\boldsymbol{x}) = (\varphi_{\Delta t/2}^{[1]} \circ \varphi_{\Delta t}^{[2]} \circ \varphi_{\Delta t/2}^{[1]})(\boldsymbol{x}). \tag{12}$$

We independently generate the initial values for the states from the standard normal distribution. We choose the input signal $y_2(t)$ from the channel F4C4 of the healthy subject H9 in the above EEG dataset which contains the EEG measurements for 18 pediatric subjects. Using the discretisation step size of $h^{\mathrm{sim}} = 1/256$, we obtain the approximate solution paths for subsystems (10) and (11) over time-length $T = 20$ seconds. We then calculate the output signal $y_1(t) = x_1(t) - x_2(t)$.

Under the above simulation setting, we first demonstrate that the ccDCM does have bifurcation behavior when $A/B$, $C$, $K$, and $v_0$ take values in some ranges. In the analysis, we change the value of one parameter, fixed other parameters to their typical values as specified in Table 1. Figure S4 respectively display the input time series from the channel F4C4 of the healthy subject H9 and the corresponding output time series under varying values of $A/B$, $C$, $K$, and $\nu_0$.

As shown in Figure S4(a), the output $y_1(t)$ remains stable when $A/B = 2/25$ and $3.25/22$. However, when $A/B$ increases to $4/14$, the output $y_1(t)$ exhibits unstable

oscillatory behavior after approximately 15 seconds. Note that, as depicted in Figure S4(a), the input signal reaches its maximum around this time. In the remaining panels of Figure S4, no similar phenomenon has been observed for the other parameters. Furthermore, the higher values of $B_A = A/B$, $K_A = AK$, $1/C_A = 1/(AC)$ and $\nu_0$, the higher output signals will result.

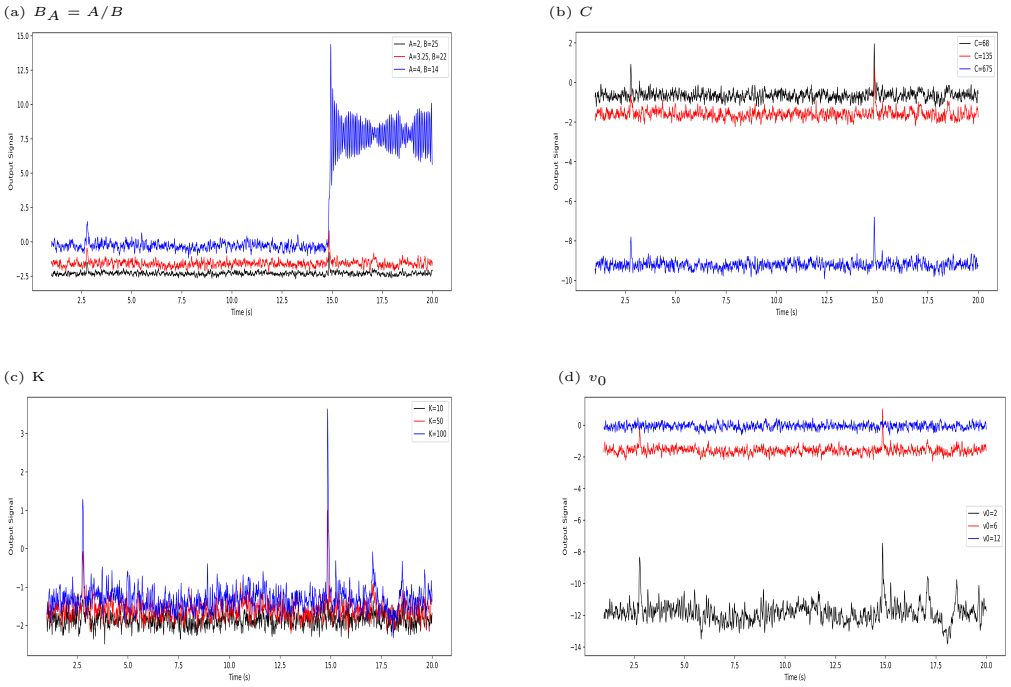



**Figure S4.** Given the input time series in Figure Output time-series for varying values of $(A, B) \in \{(2, 15), (3.25, 22), (4, 14)\}$, $C \in \{68, 135, 675\}$, $K \in \{10, 50, 100\}$ and $v_0 \in \{2, 6, 12\}$. Output time-series exhibit bifurcation behavior only when $A/B$ increases to 0.2857. The magnitude of output signal increases quickly in $A/B$, $v_0$ and $1/C$ but slowly in $K$.

## S12 Unidirectional circle network of three nodes

This section contains the mathematical expression of multi-coupled Jansen-Rit model, the simulation results for Chen-Fliess, RK and EM methods based on different initial conditions, and the the analysis of the bidirectional coupling configuration.

First, we give the mathematical expression of multi-coupled Jansen-Rit model:

$$
\begin{aligned}
dx_{A0}^{(i)}(t) &= x_{A3}^{(i)}(t)\,dt, \quad dx_1^{(i)}(t) = x_4^{(i)}(t)\,dt, \quad dx_2^{(i)}(t) = x_5^{(i)}(t)\,dt, \\
dx_{A3}^{(i)}(t) &= \left[a\,\mathrm{S}^{(i)}\left(y^{(i)}(t)\right) - 2a\,x_{A3}^{(i)}(t) - a^2\,x_{A0}^{(i)}(t)\right]dt, \\
dx_4^{(i)}(t) &= \left\{a\left[\sum_{i\neq j} K_A^{(i|j)}\,\mathrm{S}^{(i)}\left(y^{(j)}(t)\right) + 0.8C_A^{(i)}\,\mathrm{S}^{(i)}\left(C_A^{(i)}\,x_{A0}^{(i)}(t)\right)\right]\right. \\
&\quad \left. -2a\,x_4^{(i)}(t) - a^2\,x_1^{(i)}(t)\right\}dt + a\sigma_A^{(i|j)}d\tilde{\mathbf{w}}^{(i|j)}(t), \\
dx_5^{(i)}(t) &= \left[(1/B_A^{(i)})b0.25\,C_A^{(i|j)}\,\mathrm{S}^{(i)}\left(0.25C_A^{(i|j)}\,x_{A0}^{(i)}(t)\right) - 2b\,x_5^{(i)}(t) - b^2\,x_2^{(i)}(t)\right]dt, \\
y^{(i)}(t) &= x_1^{(i)}(t) - x_2^{(i)}(t).
\end{aligned}
\tag{13}
$$

with

$$
\mathrm{S}_i(x(t)) = \frac{\nu_{\max}}{1+\exp\left(r(\nu_0^{(i)} - x(t))\right)}, \quad t\in[0,T], \quad i = 1,2,3.
$$

For the unidirectional configuration shown in the Figure 13, all connections other than $K^{(2|1)}$, $K^{(3|2)}$, and $K^{(1|3)}$ were set to zero.

The simulation results for Chen-Fliess, RK and EM methods based on different initial conditions can be found in Table S13

# S13 Bidirectionally coupled system with multiple inputs

In this section, we consider a bidirectionally coupled configuration in which a single Jansen–Rit node receives inputs from multiple nodes. The mathematical formulation of this configuration is identical to that in Equation (13), and its schematic illustration is provided in Figure 13 in the main text.

The loss function for the bidirectionally coupled system differs from that for the single-input system because it incorporates multiple inputs. For the output signal $y(t) = x_1(t) - x_2(t)$, we construct the likelihood using the same approach as in the single-input case. For subject $n$ in group $k$, the resulting negative log-likelihood at node $i$ during period $q$, with inputs from nodes $j = 1,\ldots,J$, is given by

$$
L_{nq}^{(i|j)k}(\boldsymbol{\theta}_{nq}^{(i|j)k}) = \frac{D}{2}\log 2\pi + \frac{1}{2}\sum_{d=1}^{D}\log G_{nqd}^{(i|j)k} + \frac{1}{2}\sum_{d=1}^{D}\frac{\left(y_{nq}^{(i)k}(t_{b_q-1}^d) - \widehat{y}_{nq}^{(i)k}(t_{b_q-1}^d)\right)^2}{G_{nqd}^{(i|j)k}},
$$

where $G_{nqd}^{(i|j)k} = (\sigma_{Anq}^{(i)k}a)^2(t_{b_q-1}^d - t_0^d)^3/6$ and

$$
\widehat{y}_{nq}^{(i)k}(t_{b_q-1}^d) \approx g\left(t_{b_q-1}^d | x_{-\{A3\}nq}^{(i)k}(t_0^d), y_{nq}^{(i)k}(t_0^d), \dot{y}_{nq}^{(i)k}(t_0^d), \{y_{nq}^{(j)k}(t_0^d)\}_{j\neq i}^{J}, \boldsymbol{\theta}_{nq}^{(i)k}\right)
$$

**Table S13.** Median ARE values for CF, EM, and RK based methods under different initial conditions. NMADs are reported in parentheses.

| Parameter | Node | CF-based | | | EM-based | | | RK-based | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Zero | Constant | Normal | Zero | Constant | Normal | Zero | Constant | Normal |
| $B_A$ | 1 | 0.05 (0.04) | 0.05 (0.04) | 0.04 (0.03) | 0.40 (0.18) | 0.44 (0.19) | 0.46 (0.22) | 0.55 (0.17) | 0.56 (0.15) | 0.52 (0.20) |
| | 2 | 0.02 (0.02) | 0.02 (0.02) | 0.02 (0.02) | 0.80 (0.85) | 0.75 (0.38) | 0.76 (0.76) | 0.77 (0.90) | 0.71 (0.56) | 0.79 (0.84) |
| | 3 | 0.03 (0.02) | 0.02 (0.02) | 0.03 (0.03) | 0.18 (0.19) | 0.16 (0.14) | 0.14 (0.14) | 0.16 (0.17) | 0.20 (0.24) | 0.13 (0.13) |
| $C_A$ | 1 | 0.03 (0.02) | 0.03 (0.02) | 0.03 (0.03) | 0.39 (0.09) | 0.43 (0.10) | 0.42 (0.11) | 0.42 (0.13) | 0.45 (0.13) | 0.42 (0.12) |
| | 2 | 0.04 (0.03) | 0.04 (0.03) | 0.04 (0.03) | 0.97 (0.04) | 0.96 (0.05) | 0.97 (0.04) | 0.96 (0.05) | 0.96 (0.05) | 0.96 (0.05) |
| | 3 | 0.03 (0.03) | 0.02 (0.02) | 0.02 (0.02) | 0.34 (0.08) | 0.33 (0.07) | 0.32 (0.09) | 0.34 (0.16) | 0.33 (0.17) | 0.31 (0.14) |
| $\nu_0$ | 1 | 0.03 (0.02) | 0.02 (0.02) | 0.02 (0.02) | 0.09 (0.06) | 0.09 (0.07) | 0.11 (0.08) | 0.08 (0.07) | 0.08 (0.07) | 0.08 (0.07) |
| | 2 | 0.02 (0.02) | 0.02 (0.01) | 0.02 (0.02) | 0.20 (0.19) | 0.25 (0.23) | 0.17 (0.17) | 0.26 (0.20) | 0.33 (0.23) | 0.23 (0.25) |
| | 3 | 0.02 (0.02) | 0.01 (0.01) | 0.02 (0.02) | 0.20 (0.10) | 0.21 (0.07) | 0.22 (0.09) | 0.19 (0.09) | 0.20 (0.08) | 0.18 (0.09) |
| $K_A$ | 1 | 0.09 (0.07) | 0.07 (0.06) | 0.07 (0.04) | 0.34 (0.18) | 0.38 (0.18) | 0.39 (0.21) | 0.55 (0.21) | 0.58 (0.25) | 0.54 (0.25) |
| | 2 | 0.11 (0.12) | 0.13 (0.12) | 0.13 (0.09) | 0.32 (0.06) | 0.32 (0.05) | 0.31 (0.07) | 0.31 (0.06) | 0.31 (0.06) | 0.31 (0.07) |
| | 3 | 0.05 (0.04) | 0.05 (0.04) | 0.05 (0.04) | 0.10 (0.07) | 0.09 (0.07) | 0.10 (0.07) | 0.11 (0.12) | 0.11 (0.11) | 0.10 (0.09) |
| $\sigma_A$ | 1 | 0.02 (0.02) | 0.03 (0.02) | 0.02 (0.02) | 0.84 (0.01) | 0.84 (0.01) | 0.84 (0.01) | 0.83 (0.02) | 0.84 (0.01) | 0.84 (0.02) |
| | 2 | 0.10 (0.04) | 0.11 (0.04) | 0.11 (0.05) | 0.86 (0.02) | 0.86 (0.02) | 0.86 (0.02) | 0.85 (0.03) | 0.84 (0.05) | 0.85 (0.03) |
| | 3 | 0.03 (0.03) | 0.03 (0.03) | 0.04 (0.02) | 0.82 (0.01) | 0.82 (0.01) | 0.82 (0.02) | 0.82 (0.02) | 0.82 (0.03) | 0.82 (0.02) |

*Note.* Zero denotes all-zero initial conditions; Constant denotes the fixed initial condition [0, 0.5, 1, 1.5, 2, 2.5]; Normal denotes initial conditions drawn from $\mathcal{N}(0, 1)$.

with

$$
\begin{aligned}
&g(t|x^{(i)}_{-\{3\}}(t_0), y^{(i)}(t_0), \dot{y}^{(i)}(t_0), \{y^{(j)}(t_0)\}^{J}_{j\neq i}, \boldsymbol{\theta}) \\
&\quad = y^{(i)}(t_0) + \dot{y}^{(i)}(t_0)(t-t_0) \\
&\qquad + \left( \frac{\nu_{\max} a 0.8 C_A}{1 + e^{r(v_0 - C_A x^{(i)}_{A0}(t_0))}} + \sum_{i\neq j}^{J} \frac{\nu_{\max} a K_A^{(i|j)}}{1 + e^{r(v_0 - y^{(j)}(t_0))}} - a^2 x_1^{(i)}(t_0) - 2a x_4^{(i)}(t_0) \right) \\
&\qquad \times \frac{(t-t_0)^2}{2} \\
&\qquad - \left( \frac{\nu_{\max} B_A^{-1} b 0.25 C_A}{1 + e^{r(v_0 - 0.25 C_A x^{(i)}_{A0}(t_0))}} - 2b x_5^{(i)}(t_0) - b^2 x_2^{(i)}(t_0) \right) \frac{(t-t_0)^2}{2}.
\end{aligned}
$$

and $\boldsymbol{\theta} = (B_A, C_A, v_0, \sigma_A, K_A^{(i|j\neq i)})$. Furthermore, the hidden states $x_1$, $x_2$, and $x_3$ are recovered using the same procedure as that employed in the single-input case.

To investigate both the influence of multiple inputs on the proposed estimation method originally developed for the single-input case and the performance of the proposed estimation method designed for the multiple-input setting, we conducted a simulation study.

In this study, a four-node network is considered, and the inter-channel coupling structure is defined as follows: node 1 receives input from $y_4(t)$, node 2 receives input from $y_4(t)$, node 3 receives input from $y_4(t)$, and node 4 receives inputs from $y_1(t)$, $y_2(t)$, and $y_3(t)$. Consequently, node 4 acts as an integration node that aggregates information from the other three nodes, thereby forming a more complex coupling structure than that in the single-input case.

Such a configuration enables us to evaluate the robustness and adaptability of the proposed estimation framework under multi-source interactions and to assess its capability in characterising complex neural connectivity patterns commonly observed in large-scale brain networks.

In the simulation study, we set the parameters as follows: for node 1, $A_1 = 3.25$, $B_1 = 22$, $C_1 = 135$, $v_{0,1} = 6$, $K^{(1|2)} = 0$, $K^{(1|3)} = 0$, $K^{(1|4)} = 100$, and $\sigma_1 = 10$; for node 2, $A_2 = 4$, $B_2 = 14$, $C_2 = 135$, $v_{0,2} = 4$, $K^{(2|1)} = 0$, $K^{(2|3)} = 0$, $K^{(2|4)} = 50$, and $\sigma_2 = 10$; for node 3, $A_3 = 3.25$, $B_3 = 18$, $C_3 = 135$, $v_{0,3} = 5$, $K^{(3|1)} = 0$, $K^{(3|2)} = 0$, $K^{(3|4)} = 50$, and $\sigma_3 = 10$; and for node 4, $A_4 = 5$, $B_4 = 22$, $C_4 = 135$, $v_{0,4} = 6$, $K^{(4|1)} = 50$, $K^{(4|2)} = 100$, $K^{(4|3)} = 50$, and $\sigma_4 = 10$. We repeated the simulation 200 times. The simulation results were summarised in Tables S14 and S15 presented the ARE rates of the multiple-inputs adjusted method and the original single input method for estimating these unknown parameters. The results demonstrated that the multiple-inputs adjusted method outperformed the original single-input method.

Note that when the true $K_A^{(j|\cdot)} = 0$, the denominator of ARE would be zero. So, the ARE could not be useful for evaluating the error rate of estimating $K_A^{(j|\cdot)}$. Instead, for the multiple-input method, we used the mean absolute errors for estimating $K_A(j|\cdot)$ at nodes $j = 1, 2$, and 3, with NMAD values in parentheses. We obtained the mean errors of 4.25 (6.3) for $K_A^{(1|2)}$ and 1.95 (2.89) for $K_A^{(1|3)}$; 7.53 (11.12) for $K_A^{(2|1)}$ and 0 (0) for $K_A^{(2|3)}$; 25.55 (8.89) for $K_A^{(3|1)}$ and 2.68 (3.97) for $K_A^{(3|2)}$. Similarly, we estimated the errors for the single input method. The further details were omitted but they also demonstrated that the multiple-inputs adjusted method outperformed the original single-input method.

**Table S14.** The median (upper row) and the NMAD (lower row) of ARE for the proposed multiple-input estimation method.

| Node (j) | $B_A^{(j)}$ | $C_A^{(j)}$ | $v_0^{(j)}$ | $\sigma_A^{(j)}$ | $K_A^{(j\|1)}$ | $K_A^{(j\|2)}$ | $K_A^{(j\|3)}$ | $K_A^{(j\|4)}$ |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.18 | 0.04 | 0.09 | 0.43 | | $\infty$ | $\infty$ | 0.07 |
| | 0.03 | 0.02 | 0.01 | 0.01 | | $\infty$ | $\infty$ | 0.02 |
| 2 | 0.06 | 0.04 | 0.11 | 0.45 | $\infty$ | | $\infty$ | 0.04 |
| | 0.01 | 0.01 | 0.01 | 0.01 | $\infty$ | | $\infty$ | 0.04 |
| 3 | 0.15 | 0.04 | 0.08 | 0.45 | $\infty$ | $\infty$ | | 0.03 |
| | 0.03 | 0.02 | 0.01 | 0.01 | $\infty$ | $\infty$ | | 0.03 |
| 4 | 0.18 | 0.04 | 0.08 | 0.43 | 0.11 | 0.13 | 0.08 | |
| | 0.03 | 0.02 | 0.01 | 0.01 | 0.07 | 0.02 | 0.07 | |

**Table S15.** Median ARE values for the four-node system using the proposed single-input estimation method. NMAD values are reported in parentheses. Rows correspond to output nodes and columns to input nodes.

| Parameter | Output node | Input node | | | |
|---|---|---|---|---|---|
| | | **1** | **2** | **3** | **4** |
| $B_A$ | 1 | | 0.50 (0.09) | 1.17 (0.08) | 0.04 (0.03) |
| | 2 | 0.12 (0.02) | | 0.20 (0.02) | 0.03 (0.02) |
| | 3 | 0.11 (0.04) | 0.21 (0.05) | | 0.03 (0.02) |
| | 4 | 0.28 (0.03) | 0.35 (0.05) | 0.48 (0.05) | |
| $C_A$ | 1 | | 0.22 (0.06) | 0.30 (0.07) | 0.03 (0.02) |
| | 2 | 0.11 (0.02) | | 0.09 (0.03) | 0.03 (0.02) |
| | 3 | 0.07 (0.02) | 0.15 (0.04) | | 0.02 (0.02) |
| | 4 | 0.19 (0) | 0.03 (0.03) | 0.19 (0) | |
| $v_0$ | 1 | | 0.09 (0.04) | 0.22 (0.05) | 0.02 (0.02) |
| | 2 | 0.05 (0.02) | | 0.06 (0.02) | 0.01 (0.01) |
| | 3 | 0.02 (0.02) | 0.07 (0.03) | | 0.01 (0.01) |
| | 4 | 0.07 (0.02) | 0.05 (0.02) | 0.11 (0.03) | |
| $K_A$ | 1 | | $\infty$ | $\infty$ | 0.07 (0.05) |
| | 2 | $\infty$ | | $\infty$ | 0.05 (0.04) |
| | 3 | $\infty$ | $\infty$ | | 0.03 (0.03) |
| | 4 | 1.40 (0) | 0.20 (0) | 1.19 (0.28) | |
| $\sigma_A$ | 1 | | 0.57 (0.04) | 0.57 (0.06) | 0.19 (0.03) |
| | 2 | 0.05 (0.02) | | 0.12 (0.03) | 0.08 (0.02) |
| | 3 | 0.09 (0.03) | 0.19 (0.03) | | 0.04 (0.03) |
| | 4 | 0.46 (0.03) | 0.21 (0.03) | 0.70 (0.02) | |

## References


Leimkuhler, B., C. Matthews, and G. Stoltz (2016). The computation of averages from equilibrium and nonequilibrium Langevin molecular dynamics. *IMA Journal of Numerical Analysis 36*, 13–79.

Merkt, B., J. Timmer, and D. Kaschek (2015). Higher-order lie symmetries in identifiability and predictability analysis of dynamic models. *Physical Review E 92*.

Milstein, G. N. and M. V. Tretyakov (2003). Quasi-symplectic methods for Langevin-type equations. *IMA Journal of Numerical Analysis 23*, 593–626.

Yates, J. W., N. D. Evans, and M. J. Chappell (2009). Structural identifiability analysis via symmetries of differential equations. *Automatica 45*(11), 2585–2591.

Zhang, J. and A. Sanderson (2009). JADE: Adaptive Differential Evolution with Optional External Archive. *IEEE Transactions on Evolutionary Computation 13*, 945–958.